Siddhi Sudesh Sawant

Data Engineer

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SUMMARY:

- Experienced and results-driven Data Engineer with a proven track record of over 4 years of experience software development experience with expertise in Big Data, Hadoop Ecosystem, Cloud Engineering, Data Warehousing.
- Sound Experience with AWS services like Amazon EC2, S3, EMR, Amazon RDS, VPC, Amazon Elastic Load Balancing, IAM, Auto Scaling, Cloud Front, CloudWatch, and Lambda to trigger resources.
- Experience in building data pipelines using Azure Data Factory, Azure Databricks, and loading data to Azure Data Lake, Azure SQL Database, Azure SQL Data Warehouse to control and grant database access.
- Good experience with Azure services like HDInsight, Stream Analytics, Active Directory, Blob Storage, Cosmos DB, Storage Explorer.
- Strong Hadoop and platform support experience with all the entire suite of tools and services in major Hadoop Distributions – Cloudera, Amazon EMR, Azure HDInsight, and Hortonworks.
- Strong familiarity with GCP components, Google Container Builders, and client libraries, leveraging cloud SDKs for development efficiency.
- Led Proof of Concept (POC) initiatives to assess cloud offerings, including Google Cloud Platform.
- Hands-on expertise in GCP tools such as Cloud Shell SDK for configuring services like Data Proc, Storage, and BigQuery, ensuring optimal performance and scalabilit
- Proficient in handling and ingesting terabytes of Streaming data (Kafka, Spark streaming, Strom), Batch Data, Automation and Scheduling (Oozie, Airflow).
- Profound knowledge in developing production-ready Spark applications using Spark Components like Spark SQL, DataFrames, Datasets, Spark-ML and Spark Streaming.
- Expertise in developing multiple confluent Kafka Producers and Consumers to meet business requirements. Store the stream data to HDFS and process it using Spark.
- Strong working experience with SQL and NoSQL databases (Cosmos DB, MongoDB, HBase, Cassandra), data modeling, tuning, disaster recovery, backup and creating data pipelines.
- Experienced in scripting with Python (PySpark), Scala and Spark-SQL for development, aggregation from various file formats such as XML, JSON, CSV, Parquet.
- Great experience in data analysis using HiveQL, Hive-ACID tables, Pig Latin queries, custom MapReduce programs and achieved improved performance.
- Experience in monitoring document growth and estimating storage size for large MongoDB clusters as part of the data life cycle management.
- Hands-on experience on Ad-hoc queries, Indexing, Replication, Load balancing, Aggregation in MongoDB.
- Expertise in creating Kubernetes cluster with cloud formation templates and PowerShell scripting to automate deployment in a cloud environment.
- Sound knowledge in developing highly scalable and resilient Restful APIs, ETL solutions, and third-party integrations as part of Enterprise Site platform using Informatica.
- Highly involved in all facets of SDLC using Waterfall and Agile Scrum methodologies.
- Involved in migration of the legacy applications to cloud platform using DevOps tools like GitHub, Jenkins, JIRA, Docker, and Slack

TECHNICAL SKILLS:

- **Programming Languages:** SQL, Pig Latin, HiveQL, Scala, Python, Unix Shell Scripting.
- **Databases:** MS-SQL SERVER, Oracle, MS-Access, MySQL, Teradata, PostgreSQL, DB2.
- **Big Data Technologies:** Yarn, MapReduce, Pig, Hive, HBase, Cassandra, Oozie, Apache Spark, Scala, Impala, Kafka.
- **Hadoop Distributions:** Apache Hadoop 2.x/1.x, Cloudera CDP, Hortonworks HDP, Amazon EMR (EMR, EC2, EBS, RDS, S3, Glue, Elasticsearch Lambda, Kinesis, SQS, DynamoDB, Redshift, ECS) Azure HDInsight (Databricks, Data Lake, Blob Storage, Data Factory, SQL DB, SQL DWH, Cosmos DB, Azure DevOps, Active Directory).
- **NoSQL Database:** Cassandra, MongoDB.
- **Reporting Tools/ETL Tools:** Informatica, Talend, SSIS, SSRS, SSAS, ER Studio, Tableau, Power BI.
- **Methodologies:** Agile/Scrum, Waterfall.
- **Development Tools:** Eclipse, NetBeans, IntelliJ, Hue, Microsoft Office Suite (Word, Excel, PowerPoint, Access)
- **Operating Systems:** Windows, Macintosh, Linux, Ubuntu, Unix.
- **Others:** Machine learning, NLP, Stream Sets, Spring Boot, Jupyter Notebook, Docker, Kubernetes, Jenkins, Jira.

EDUCATION:

Northeastern University Boston May 2024
Master in Computer Science (MSIS)
Pune University, India May 2020
Bachelor of Engineering in Electronics and Telecommunication

PROFESSIONAL EXPERIENCE:

Data Engineer - Inventory Forecast Analyst | *BCBS USA* **Sep 2024 – Current**

- Mastered a wide array of Azure technologies, including HDInsight, Databricks, Data Lake, Blob Storage, Data Factory, Synapse Analytics, Azure SQL Database, and SQL Data Warehouse, to streamline cloud operations and data management.
- Engineered scalable data pipelines using Azure Data Factory and Synapse Analytics to ensure high-quality data integration and accessibility, supporting key healthcare and insurance data initiatives.
- Led the development of automated workflows to handle daily incremental data loads from RDBMS systems to Azure Data Lake, improving data availability and reliability for healthcare analytics.
- Utilized a robust suite of Azure tools, including HDInsight, Databricks, and Blob Storage, to streamline cloud operations, data processing, and storage, aligning with compliance and efficiency standards in the healthcare industry.
- Designed and deployed custom input adapters with Spark, Sqoop, and Oozie, optimizing the ingestion of data from multiple healthcare databases into Azure Data Lake.
- Implemented ETL processes with Azure Data Factory, T-SQL, Spark SQL, and U-SQL to support seamless data movement and transformation, catering to the complex needs of healthcare data analytics.

- Developed and maintained data integration scripts in Azure Data Factory, utilizing SQL Activity for efficient data transformation, aligning with healthcare industry requirements for data accuracy and security.
- Enhanced data orchestration processes through the use of Airflow Operators and Python libraries, improving data flow and processing efficiencies critical to insurance data operations.
- Employed Databricks notebooks and Spark APIs for advanced data analysis, providing insights crucial for healthcare data-driven decision-making.
- Integrated and unified data from various sources, including MongoDB, MS SQL, and cloud platforms, using Azure Data Factory, ensuring consistent data standards across healthcare and insurance datasets.
- Created interactive Power BI dashboards to visualize healthcare and insurance data trends, offering actionable insights for stakeholders.
- Spearheaded CI/CD processes for seamless deployment of applications in Azure Cloud, strengthening system reliability and operational agility in healthcare data management environments.

Graduate Assistance | *Northeastern University, MA, USA* **Jan 2023 – Apr 2024**

- Analyzed historical student data using SQL and Python to identify key factors influencing student retention, enhancing data-driven decision-making within the university.
- Developed a predictive model using Logistic Regression and Random Forest in Scikit-learn to forecast at-risk students, improving intervention strategies for academic counselors.
- Designed and implemented an interactive Tableau dashboard to visualize real-time retention metrics, empowering university leadership to monitor student performance.
- Applied statistical methods such as hypothesis testing and correlation analysis in R to validate significant factors affecting retention, supporting evidence-based policy decisions.
- Synthesized insights from exploratory data analysis (EDA) to uncover trends and relationships between academic, financial, and demographic factors, leading to the development of targeted retention programs.
- Evaluated model performance using cross-validation and metrics like accuracy and precision, optimizing predictive model outcomes to identify students at risk of dropping out.

Software Engineer/ Data Engineer | *Capgemini Technology Service Pvt. Ltd, India* **Apr 2021 – Aug 2022**

- Contributed to diverse projects in Data Engineering and Mainframes technologies at Capgemini Technology Services Ltd.
- Utilized key components within the Hadoop ecosystem such as Spark, HDFS, Hive, Sqoop, HBase, Zookeeper, and Oozie to develop various data loading strategies and performed transformations for analyzing large datasets.
- Implemented data warehousing with Hive and seamless data load and transform from Relational Database Systems to HDFS using Sqoop.
- Fine-tuned Spark applications to improve overall processing time for data pipelines.
- Applied advanced SQL queries including Joins, Correlated Subqueries, Views, Stored Procedures, and Triggers for data analysis.
- Demonstrated good knowledge of Hadoop architecture and various components, including HDFS, Job Tracker, Task Tracker, Name Node, Data Node, and internal workings of MapReduce, Spark Batch processing, Spark Streaming, and PySpark frameworks.
- Initiated as a Mainframes Operator, showcasing expertise in Mainframes operations, including JCL, COBOL, CICS, IBM DB2, and CA7.
- Experience in consuming and processing streaming data from Kafka using Spark Streaming API and flatten the data then store the data in Data Lake in parquet format.
- Designed and implemented data processing workflows using Azure Data Factory, orchestrating the movement and transformation of data between various sources and targets.
- Leveraged Azure Databricks for large-scale data processing and analytics, utilizing its collaborative notebook environment for interactive data exploration and model development.
- Developed and optimized complex SQL queries for data analysis and transformation, ensuring efficient data retrieval and processing.
- Designed and implemented data models in Azure SQL Database, ensuring scalability, performance, and data integrity.
- Utilized Azure Blob Storage for scalable and cost-effective storage of unstructured data, implementing data lifecycle management strategies to optimize storage costs.
- Developed and deployed RESTful APIs in Azure using Azure Functions, enabling seamless integration with other applications and services.
- Engaged in the Software Development Life Cycle (SDLC) using Agile scrum methodology for Analysis, Design, Development, Implementation, Maintenance, and Support.

Data Engineer | *Trigent Software, India* **Feb 2020 – Mar 2021**

- Engineered and managed the creation of 10 automated, scalable, code-based data pipelines, leveraging Amazon Redshift, Data Grip, Amazon S3, and Glue. These pipelines efficiently processed millions of data points, demonstrating a high level of expertise in cloud-based data architecture and automation.
- Successfully executed query optimizations on AWS, utilizing Amazon RDS and Amazon Redshift, to improve database and data warehouse performance by 32%.
- Utilized AWS Athena to manipulate CSV data files stored in AWS S3, applying Scala queries for proficient data extraction and transformation, showcasing versatility in data handling methodologies.
- Formulated and implemented Python solutions to extract data from AWS S3 and populate it into SQL Server, directly supporting business team requirements through effective data integration strategies.
- Played a key role in contributing to a Databricks Delta Lake environment on AWS, employing Spark for sophisticated data processing tasks, enhancing data lake utility and performance.
- Conducted ETL operations using Python, Spark SQL, S3, and Redshift, handling substantial volumes of data to derive actionable customer insights, thus driving business intelligence initiatives.
- Orchestrated automated CI/CD pipelines utilizing AWS CodePipeline, Jenkins, and AWS CodeDeploy, significantly enhancing deployment efficiency and operational agility.
- Authored PySpark scripts to streamline ETL processes, extracting data from S3 with a crawler and generating a data catalog to consolidate metadata.

Data Analyst

- Utilized NumPy and Pandas for data preprocessing and analysis, improving data processing time and accuracy in data modeling, supporting decision-making processes.
- Executed SQL techniques to streamline data queries, enhancing overall data analysis efficiency and generating actionable insights for informed decision-making.
- Created interactive data visualizations using Tableau, effectively supporting strategic decision-making processes within the organization.
- Adopted Agile methodologies, particularly SCRUM, in analytical projects, fostering rapid iterations and continuous feedback loops, enhancing project delivery speed and adaptability to evolving requirements in data analysis processes.